여기가 가장 중요함

Chap.4 인공 신경망 (Artificial Neural Network)

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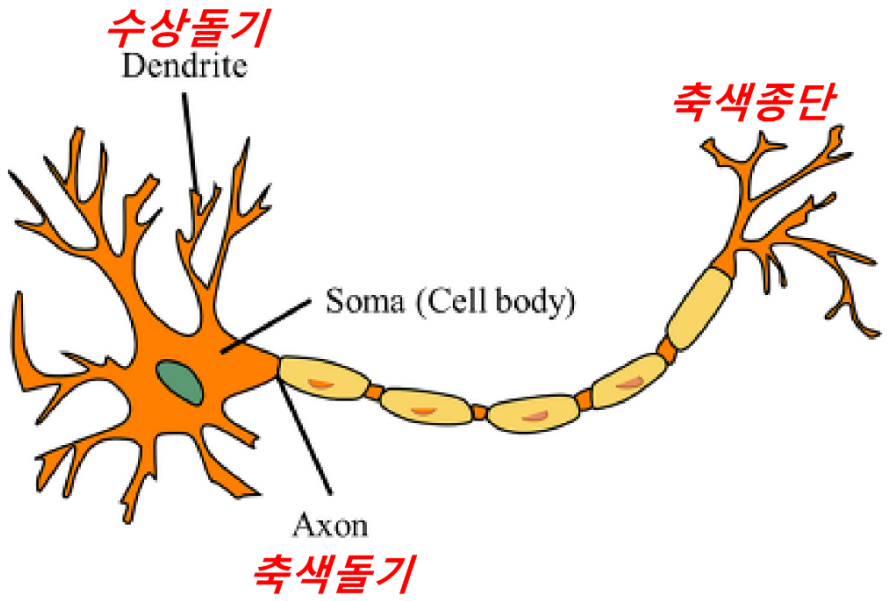
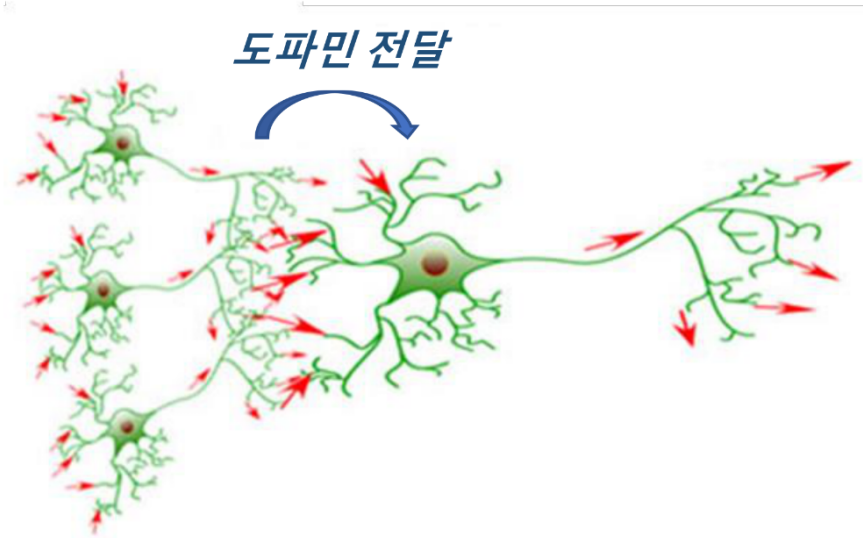
2023년도 2학기
머신러닝의 이해

신경망 (Neural Network)



Neuron

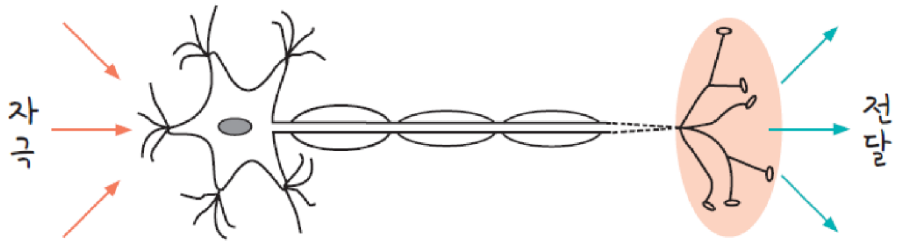


Neural Network in Brain

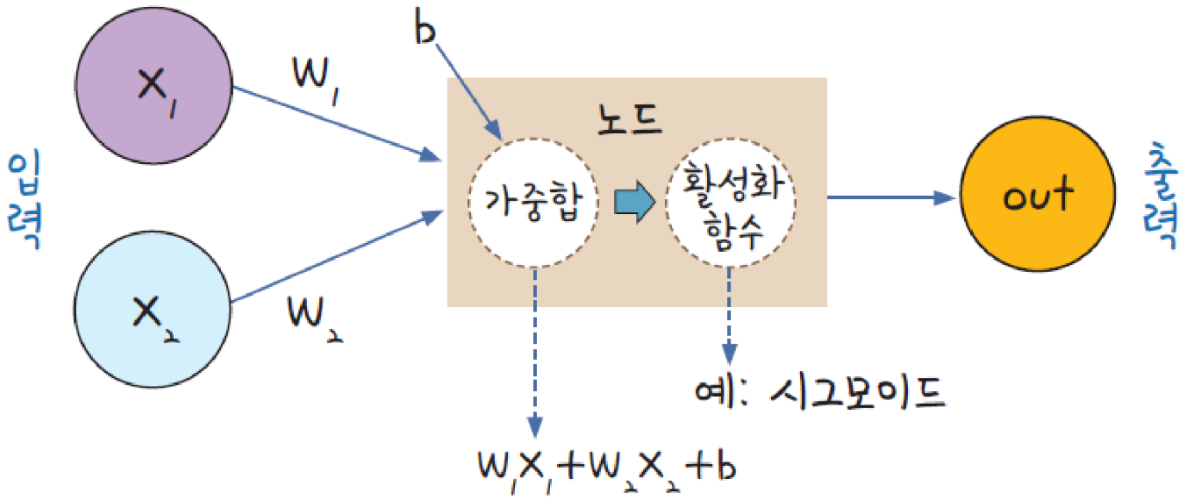
- 축색종단에서의 도파민(신경전달물질) 분비 여부
  - 전달된 신호의 강도에 따라 결정됨
  - 즉, 신호의 강도가 임계값을 넘으면 도파민 분비 ➔ **활성화**
  - 신호의 강도가 입계값을 넘지 않으면 도파민 분비되지 않음 ➔ **비활성화**

Artificial Neuron

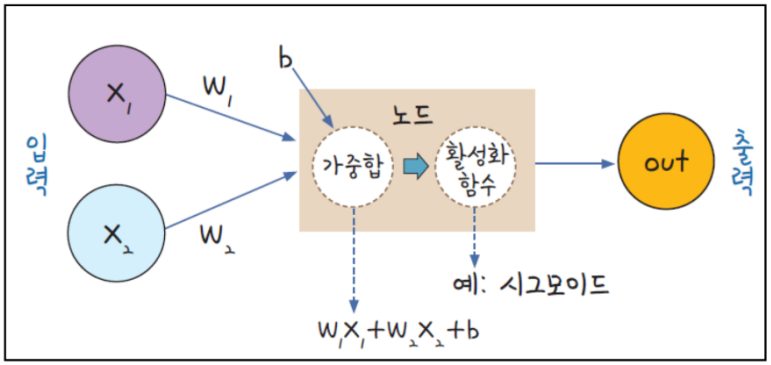
- **Neuron의 핵심 기능**
  - **전파(Propagation)** 기능
    - 신호(도파민)를 받아 전달
  - **활성화(Activation)** 기능
    - 신호의 강도에 따라 신호 활성화 유무를 결정

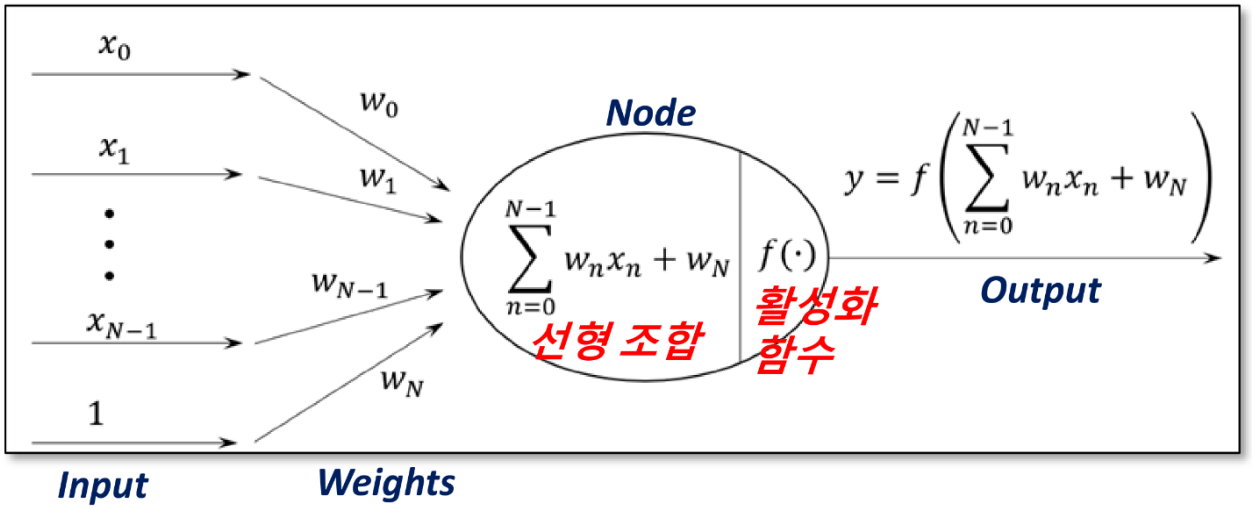


- **Artificial Neuron(인공 뉴런)**
  - Neuron 세포를 모방하여 설계



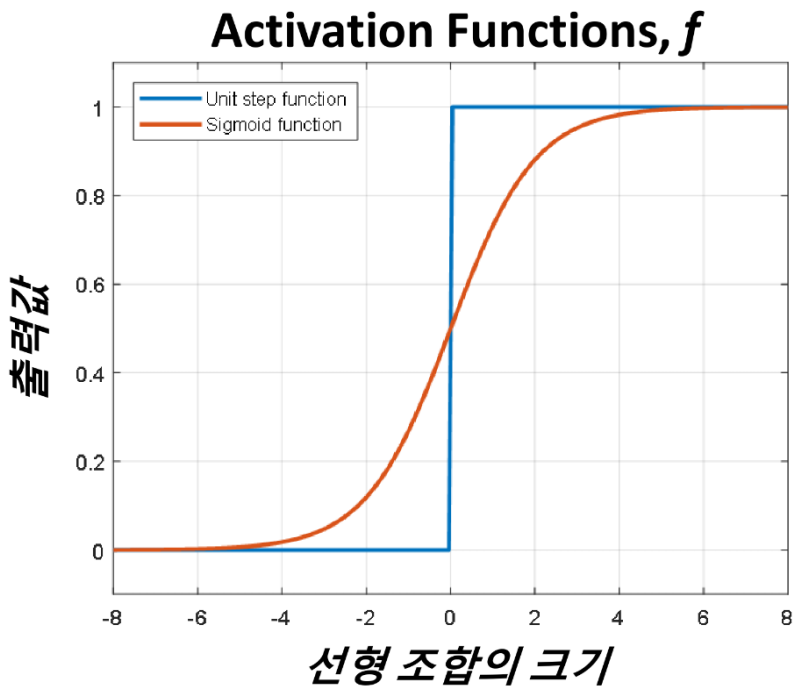
Artificial Neuron





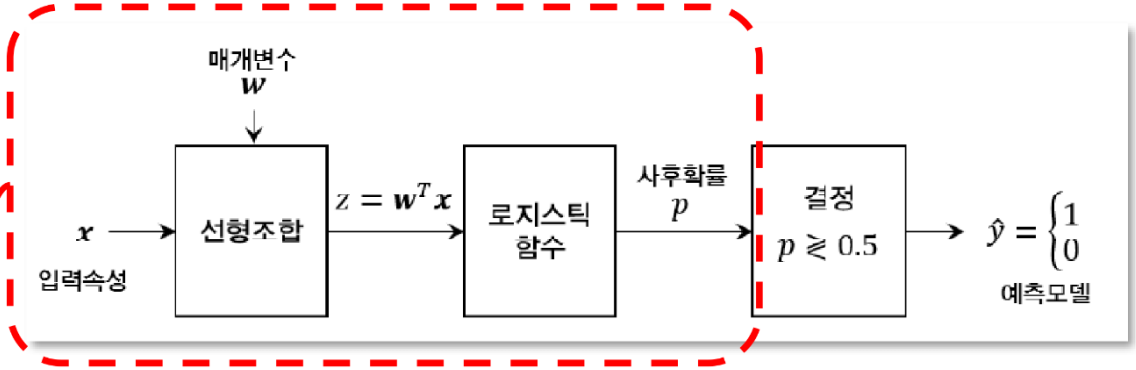
- **전파(Propagation)** 기능
  - 입력과 Weights 들의 선형 조합을 전파

- **활성화(Activation)** 기능
  - 선형 조합의 크기에 따라 다음 뉴런에 전달할 지 여부를 결정

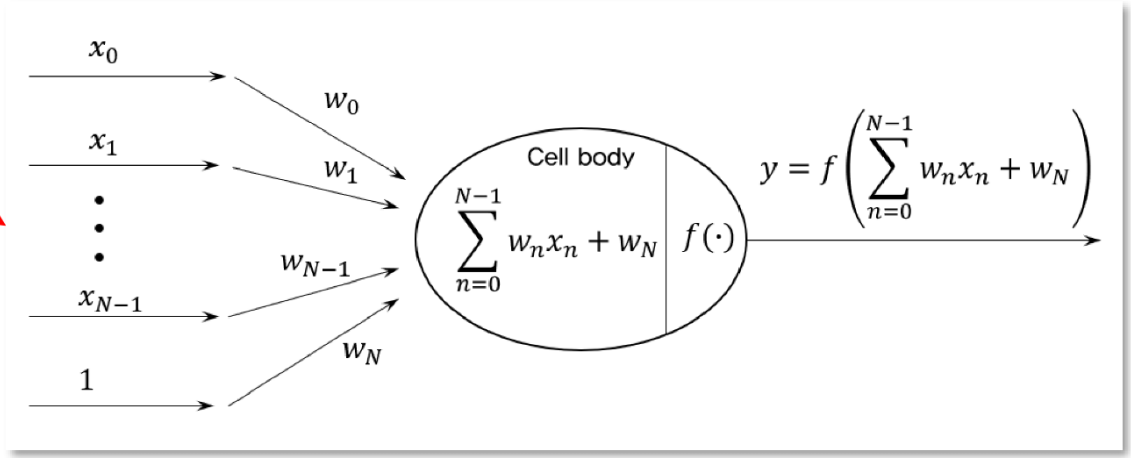


Logistic Regression과의 비교

- **Artificial Neuron의 관점에서의 Logistic Regression**
  - 일반적인 Logistic Regression 모델

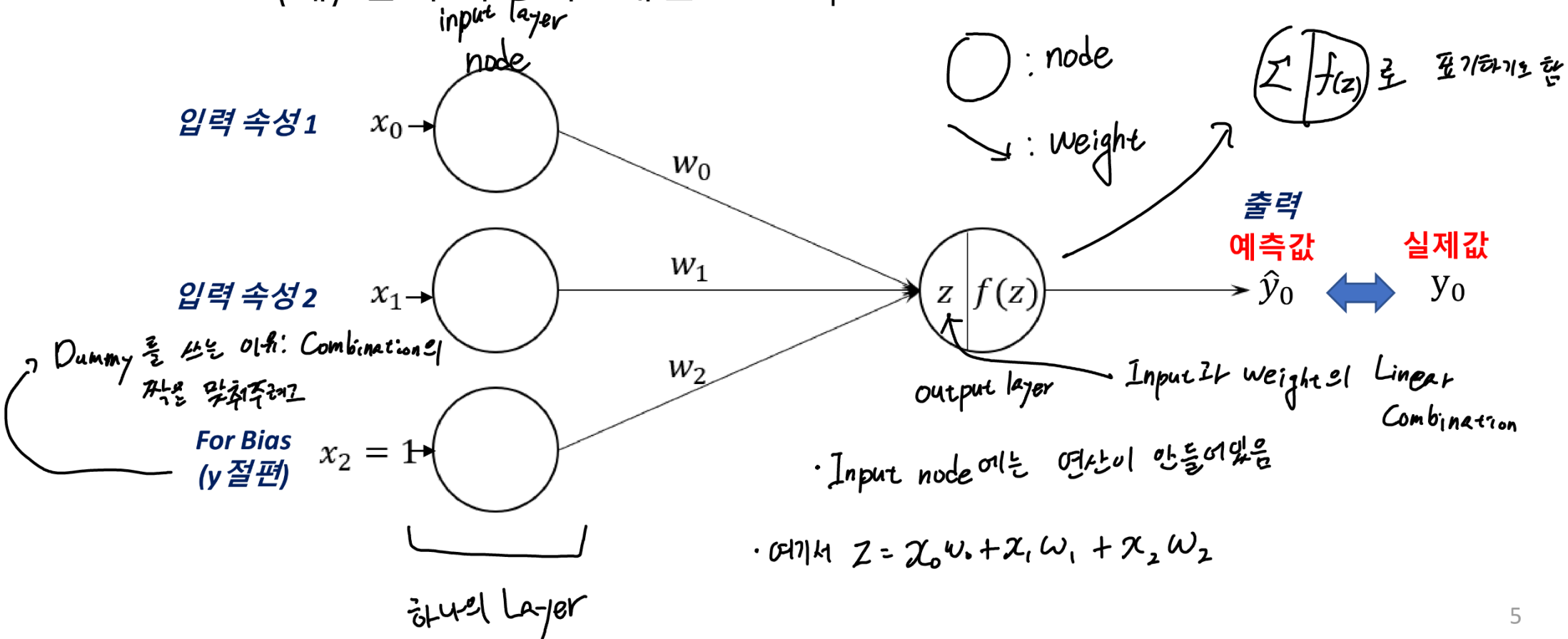


  - Sigmoid 함수를 Activation Function으로 사용하는 Artificial Neuron과 동일한 구조



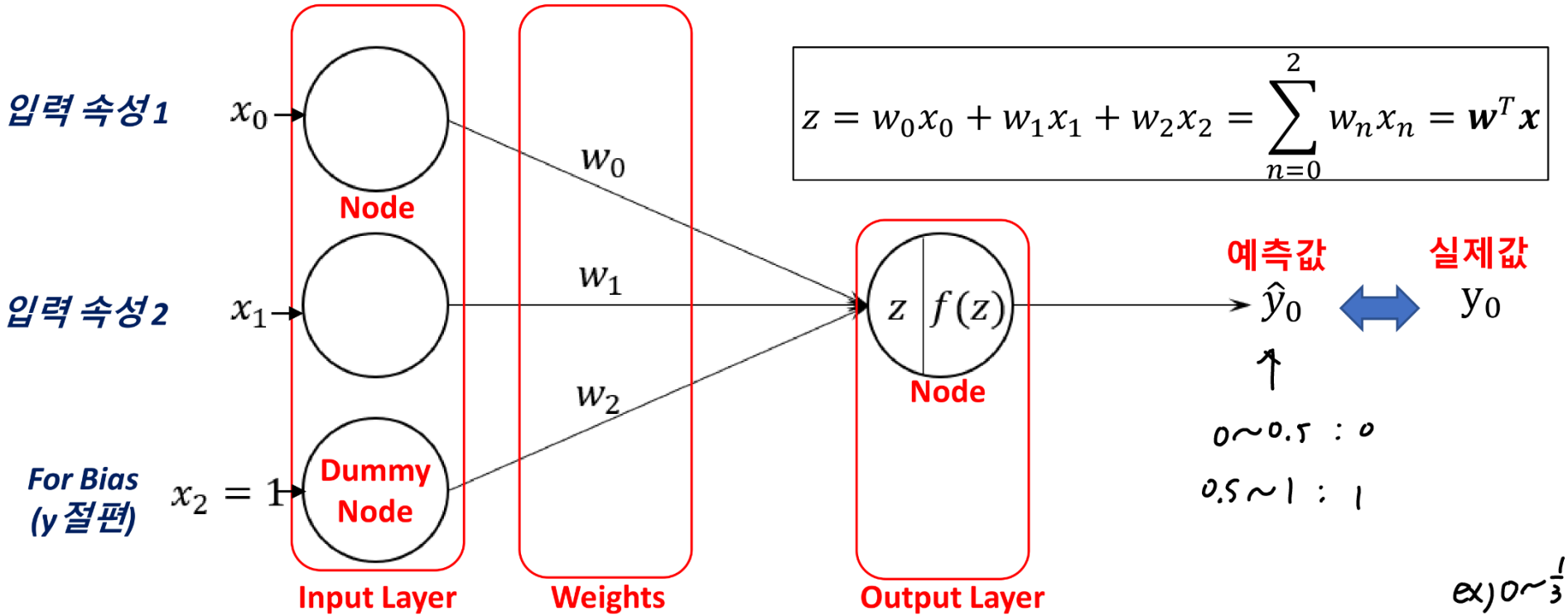
Perceptron

- **퍼셉트론 (Perceptron)**
  - 입력층과 출력층 만으로 구성된 Single-Layer Neural Network
    - Neural Network의 **기본 구성 단위**
    - Binary Classification 시스템 (출력이 0 or 1)
    - (예) 입력 속성이 2개인 Perceptron



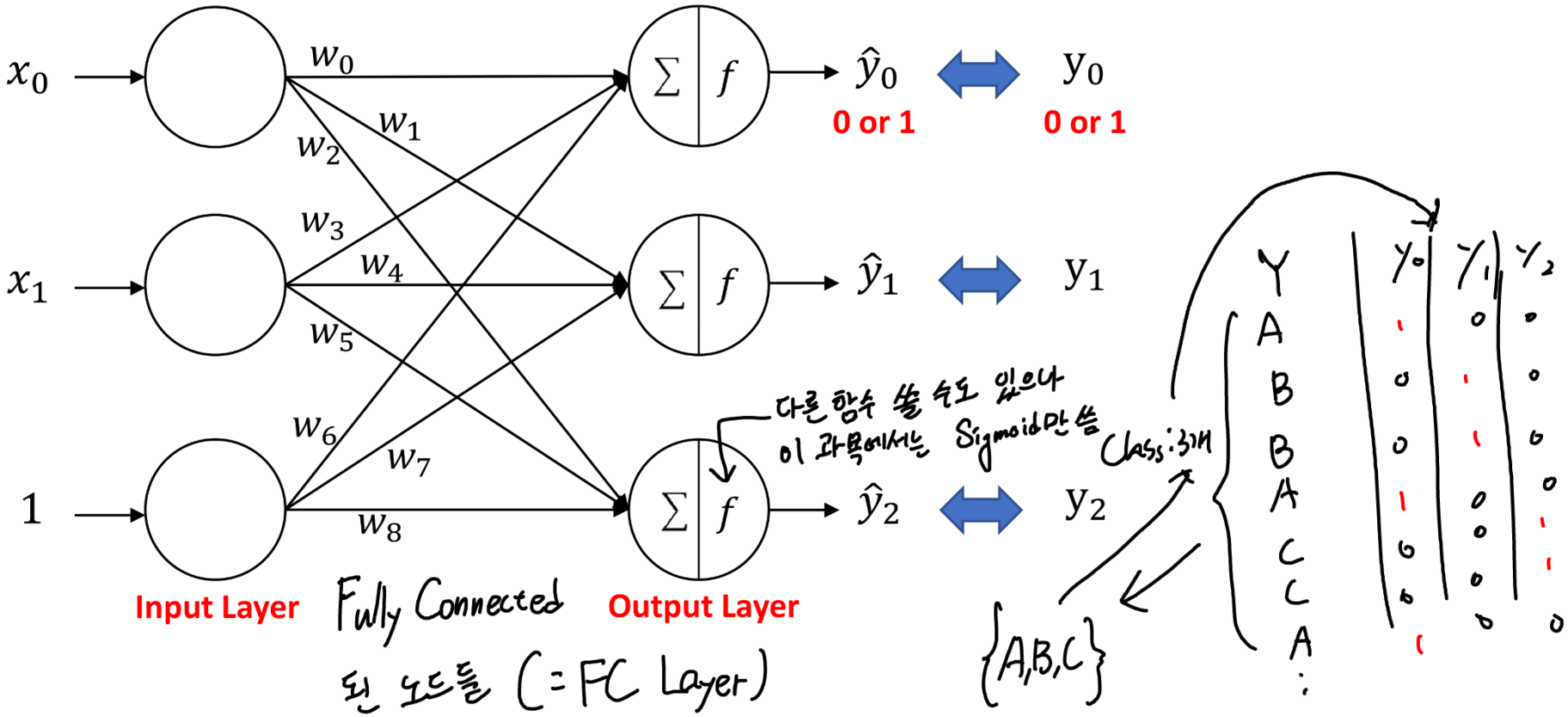
Perceptron

- **퍼셉트론 (Perceptron)**
  - Node: 해당 Node로 들어온 **모든 값을 더한 다음 Activation Function을 통과시킨 값을 출력**
    - Input node의 Activation Function: $f(x)=x$ (별도표기하지 않음)
  - 화살표: 화살표 위에 적힌 weight만큼 **곱 연산**을 수행
    - 별도 표기가 없는 화살표의 weight는 1



분류해야하는 Class가 3개 이상이라면? → 구간을 더 잘게 나누면 됨 1/3~2/3
but 너무 자잘한 단점 존재

Multi-Class Classification in Single Layer



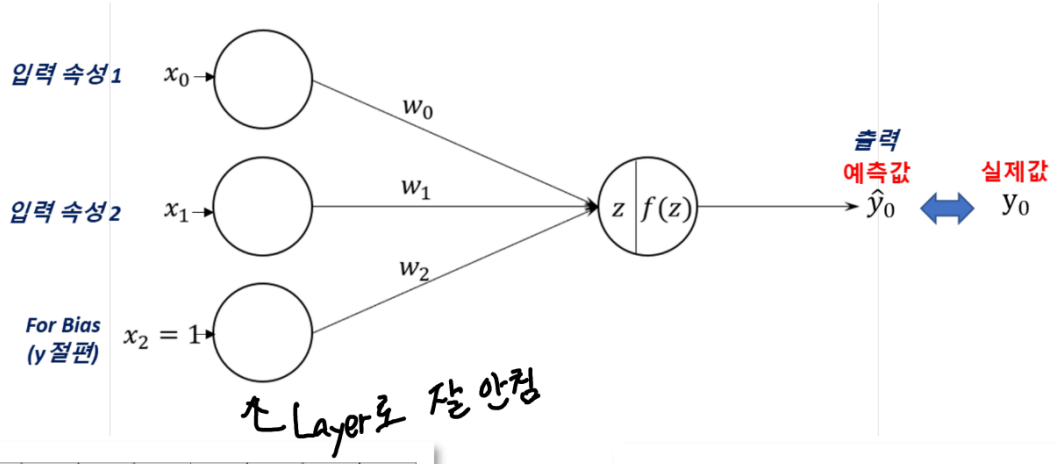
- **One-Hot Encoding (범주형 Class의 이진화)**
  - 확률론적 접근 + 기계가 인식 가능한 숫자
    - ✓ Binary Class: 음성 or 양성 => 0 or 1
    - ✓ Multi-Class: 개 or 고양이 or 말 => 100 or 010 or 001

(y_0, y_1, y_2)

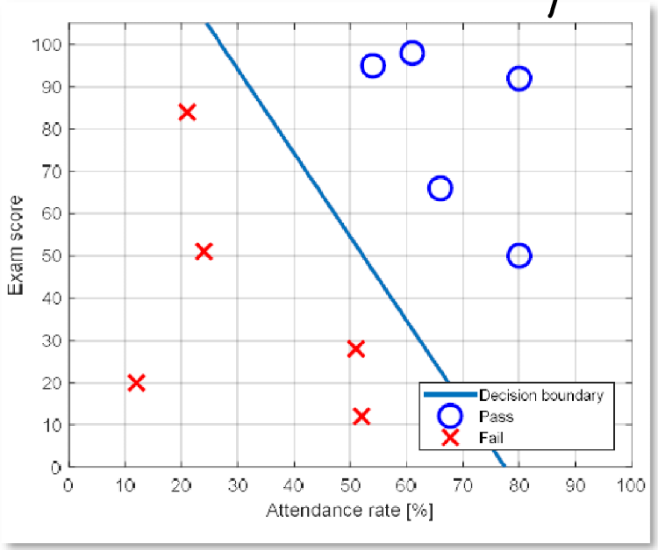
① Class 수 확인
② Class 수만큼 0으로 채움
③ 1 넣기
④ 100 (A) = t_1
010 (B) = t_2
⋮
=> Class 3개 이상의 One Hot Encoding

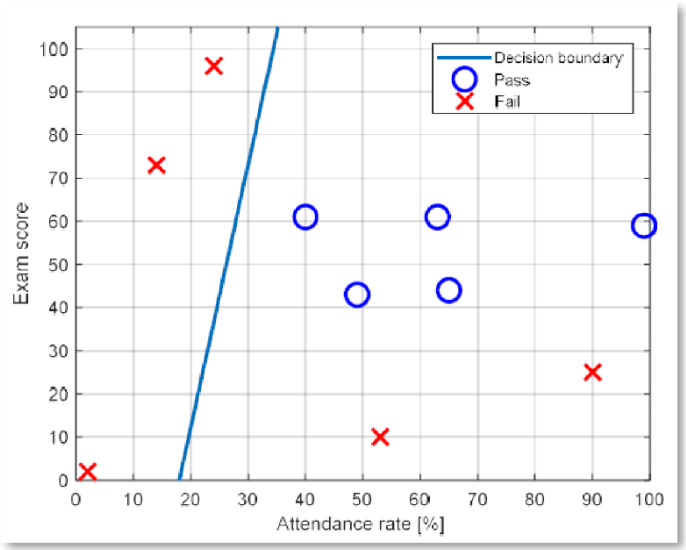
Single Layer Neural Network의 한계

- **Single Layer Neural Network의 한계**
  - 단층 신경망은 로지스틱 회귀의 성능과 동일
  - **속성 공간의 "선형 분할"만 가능**
    - **Decision Boundary가 선형적**



1-Layer로 잘 안됨

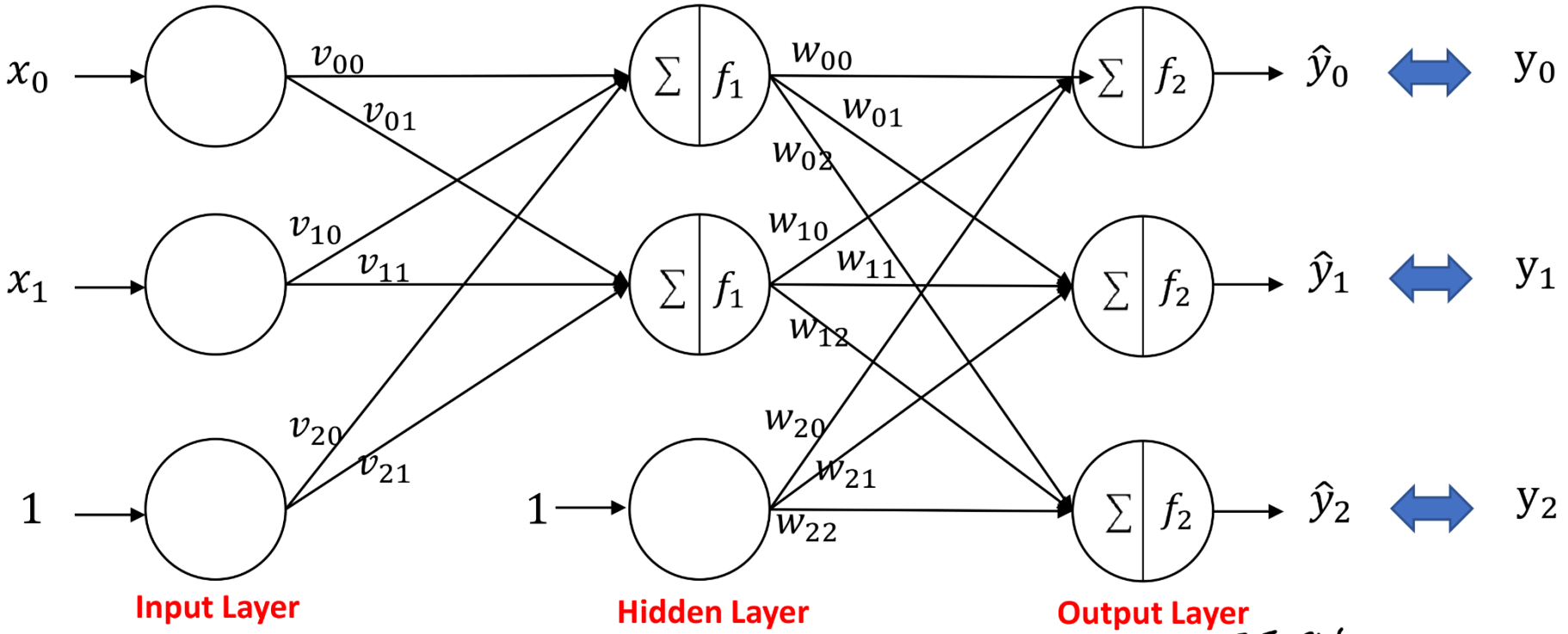




Multi-Layer Neural Network

비선형함수

- **Two-Layer Neural Network** $\hat{y}_0$: $(x_0 v_{00} + \cdots$ 를 Sigmoid에 넣은것), $\cdots$ 를 Sigmoid에 넣은 것



Neural Network 장점: 비선형적 결과 도출 가능

- **Input layer (입력층)**
  - 2 Input: 일반노드 2개 + 더미노드 1개 *<= 데이터의 Input 속성 수에 의해 결정*
- **Hidden Layer (은닉층)**
  - 일반노드 2개 + 더미노드 1개 *<= 모델 설계자가 결정 (Hyper-parameter)*

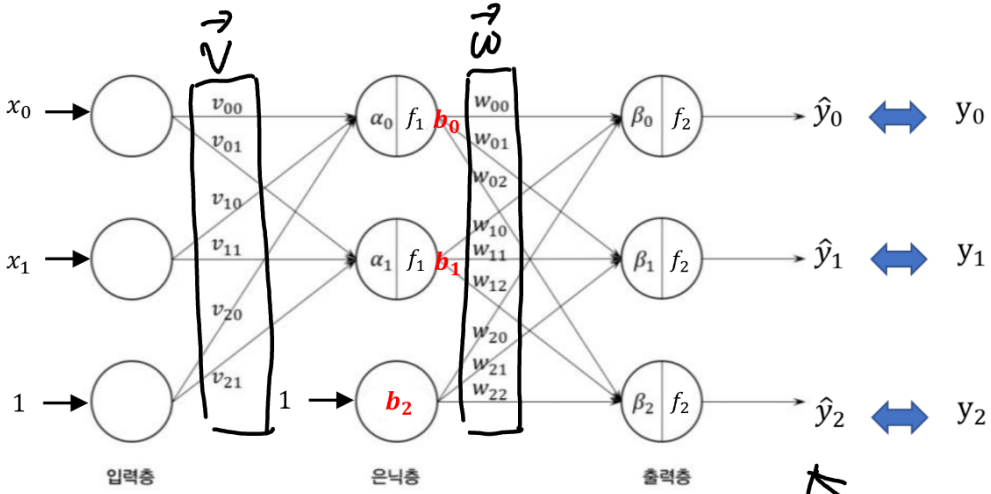
↳ 설계자가 결정하는 값

- **Output Layer (출력층)**
  - 3-Class: 일반노드 3개 *<= 데이터의 Output class 수에 의해 결정*

Multi-Layer Neural Network

- **Node 별 입출력 계산**

v_{01} : 0번째 노드 → 1번째 노드의 weighe



Hidden Layer 입출력 관계

$$\alpha_0 = v_{00}x_0 + v_{10}x_1 + v_{20}x_2$$
$$\alpha_1 = v_{01}x_0 + v_{11}x_1 + v_{21}x_2$$

$x_2 = 1$

$$b_0 = f_1(\alpha_0)$$
$$b_1 = f_1(\alpha_1) \quad b_2 = 1$$

Output Layer 입출력

$$\beta_0 = w_{00}b_0 + w_{10}b_1 + w_{20}b_2$$
$$\beta_1 = w_{01}b_0 + w_{11}b_1 + w_{21}b_2$$
$$\beta_2 = w_{02}b_0 + w_{12}b_1 + w_{22}b_2$$

$$\hat{y}_0 = f_2(\beta_0)$$
$$\hat{y}_1 = f_2(\beta_1)$$
$$\hat{y}_2 = f_2(\beta_2)$$

Input ↔ Hidden 사이의 weighe : v

- $Input \cdot \vec{v} = \alpha$
- $Sigmoid(\alpha) = b$
- $b \cdot \vec{w} = \beta$
- $Sigmoid(\beta) = y$

Artificial Neural Network(ANN) 모델의 학습 목표

$\hat{y}_0 \approx y_0, \hat{y}_1 \approx y_1, \hat{y}_2 \approx y_2$를 만족하는 최적 매개변수를 찾는다.

n → k : v_{nk} ⊕ 국제표준: n → k : v_{kn}

Two-Layer Neural Network의 일반화

2-Input, 3-Class with 2-Hidden node

Dummy node는 일반적으로 카운팅하지 않음.

Hidden Layer 입출력

$x_2 = 1$

$$\alpha_0 = v_{00}x_0 + v_{10}x_1 + v_{20}x_2$$
$$\alpha_1 = v_{01}x_0 + v_{11}x_1 + v_{21}x_2$$

$$b_0 = f_1(\alpha_0)$$
$$b_1 = f_1(\alpha_1)$$
$$b_2 = 1$$

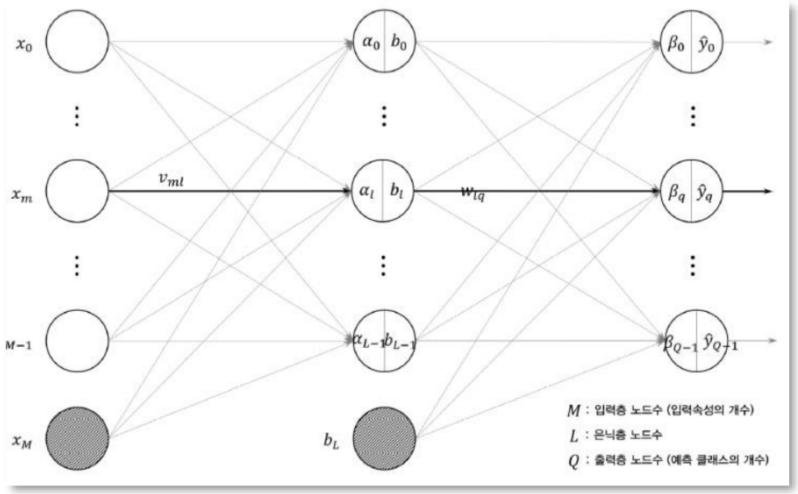
$\vec{b} = f(\vec{\alpha})$

Output Layer 입출력

$$\beta_0 = w_{00}b_0 + w_{10}b_1 + w_{20}b_2$$
$$\beta_1 = w_{01}b_0 + w_{11}b_1 + w_{21}b_2$$
$$\beta_2 = w_{02}b_0 + w_{12}b_1 + w_{22}b_2$$

$$\hat{y}_0 = f_2(\beta_0)$$
$$\hat{y}_1 = f_2(\beta_1)$$
$$\hat{y}_2 = f_2(\beta_2)$$

M-Input, Q-Class with L-Hidden node



M : 특징 수

L : 히든노드 수

Q : 클래스 수

L개

L by M+1

M+1개

$$\begin{bmatrix} \alpha_0 \\ \alpha_1 \end{bmatrix} = \begin{bmatrix} v_{00} & \cdots \\ v_{01} & \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix}$$

$\Rightarrow \vec{v} \cdot \vec{x} = \vec{\alpha}$

Two-Layer Neural Network의 일반화

2-Input, 3-Class with 2-Hidden node

Dummy node는 일반적으로 카운팅하지 않음.

Hidden Layer 입출력

$x_2 = 1$

$$\alpha_0 = v_{00}x_0 + v_{10}x_1 + v_{20}x_2$$
$$\alpha_1 = v_{01}x_0 + v_{11}x_1 + v_{21}x_2$$

$$b_0 = f_1(\alpha_0)$$
$$b_1 = f_1(\alpha_1)$$
$$b_2 = 1$$

Output Layer 입출력

$$\beta_0 = w_{00}b_0 + w_{10}b_1 + w_{20}b_2$$
$$\beta_1 = w_{01}b_0 + w_{11}b_1 + w_{21}b_2$$
$$\beta_2 = w_{02}b_0 + w_{12}b_1 + w_{22}b_2$$

$$\hat{y}_0 = f_2(\beta_0)$$
$$\hat{y}_1 = f_2(\beta_1)$$
$$\hat{y}_2 = f_2(\beta_2)$$

M-Input, Q-Class with L-Hidden node

Hidden Layer 입출력

$x_M = 1$

$$\alpha_l = v_{0l}x_0 + v_{1l}x_1 + \cdots + v_{Ml}x_M = \sum_{m=0}^{M} v_{ml}x_m$$

$$l = 0, 1, \cdots, L-1$$

$$b_l = f_1(\alpha_l) = f_1\left(\sum_{m=0}^{M} v_{ml}x_m\right) \qquad b_L = 1$$

Output Layer 입출력

$$\beta_q = w_{0q}b_0 + w_{1q}b_1 + \cdots + w_{lq}b_l + w_{Lq}b_L$$
$$= \sum_{l=0}^{L} w_{lq}b_l, \qquad q = 0, 1, \cdots, Q-1$$

$$\hat{y}_q = f_2(\beta_q) = f_2\left(\sum_{l=0}^{L} w_{lq}b_l\right)$$

Two-Layer Neural Network의 일반화

- 1개의 데이터에 대해: *M*-Input, Q-Class with *L*-Hidden node 연산의 행렬화

Hidden Layer 입력

$$\alpha_l = v_{0l}x_0 + v_{1l}x_1 + \cdots + v_{Ml}x_M = \sum_{m=0}^{M} v_{ml}x_m \qquad x_M = 1$$

$$l = 0, 1, \cdots, L-1$$

Hidden Layer 출력

$$b_l = f_1(\alpha_l) = f_1\left(\sum_{m=0}^{M} v_{ml}x_m\right) \qquad b_L = 1$$

$$b_0 = f_1(\alpha_0)$$
$$b_1 = f_1(\alpha_1)$$
$$\vdots$$
$$b_l = f_1(\alpha_l)$$
$$\vdots$$
$$b_{L'} = f_1(\alpha_{L'})$$
$$b_L = 1$$

$$\Rightarrow \quad \boldsymbol{b} = \begin{bmatrix} f_1(\alpha_0) \\ f_1(\alpha_1) \\ \vdots \\ f_1(\alpha_l) \\ \vdots \\ f_1(\alpha_{L'}) \\ 1 \end{bmatrix}$$

(L+1 by 1)

$L' = L - 1$ 일 때, (M+1개)

$$\alpha_0 = v_{00}x_0 + v_{10}x_1 + \cdots + v_{M0}x_M$$
$$\alpha_1 = v_{01}x_0 + v_{11}x_1 + \cdots + v_{M1}x_M$$
$$\vdots$$
$$\alpha_l = v_{0l}x_0 + v_{1l}x_1 + \cdots + v_{Ml}x_M$$
$$\vdots$$
$$\alpha_{L'} = v_{0L'}x_0 + v_{1L'}x_1 + \cdots + v_{ML'}x_M$$

L개

$$\Downarrow$$

α *(L by 1)* ***v*** *(L by M+1)* ***x*** *(M+1 by 1)*

$$\begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_l \\ \vdots \\ \alpha_{L'} \end{bmatrix} = \begin{bmatrix} v_{00} & v_{10} & \cdots & v_{M0} \\ v_{01} & v_{11} & \cdots & v_{M1} \\ \vdots & & \ddots & \vdots \\ v_{0l} & v_{1l} & \ddots & v_{Ml} \\ \vdots & & \ddots & \vdots \\ v_{0L'} & v_{1L'} & \cdots & v_{ML'} \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_M \end{bmatrix}$$

L개 M+1개 (dummy)

$$\Downarrow$$

$$\boldsymbol{\alpha} = \boldsymbol{v}\boldsymbol{x}$$

Two-Layer Neural Network의 일반화

- 1개의 데이터에 대해: *M*-Input, *Q*-Class with *L*-Hidden node 연산의 행렬화

Output Layer 입출력

$$\beta_q = w_{0q}b_0 + w_{1q}b_1 + \cdots + w_{lq}b_l + w_{Lq}b_L$$

$$= \sum_{l=0}^{L} w_{lq}b_l, \qquad q = 0, 1, \cdots, Q-1$$

$$\hat{y}_q = f_2(\beta_q) = f_2\left(\sum_{l=0}^{L} w_{lq}b_l\right)$$

$\vec{x}\vec{v} = \vec{\alpha}$

$\Downarrow$

$f(\vec{x}) = \vec{b}$

$\Downarrow$

$\vec{b}\vec{w} = \vec{\beta}$

$Q' = Q - 1$ 일 때,

β *(Q by 1)* ***w*** *(Q by L+1)* ***b*** (L+1 by 1)

$$\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_q \\ \vdots \\ \beta_{Q'} \end{bmatrix} = \begin{bmatrix} w_{00} & w_{10} & \cdots & w_{L0} \\ w_{01} & w_{11} & \cdots & w_{L1} \\ \vdots & & \ddots & \vdots \\ w_{0q} & w_{1q} & \ddots & w_{Lq} \\ \vdots & & \ddots & \vdots \\ w_{0Q'} & w_{1Q'} & \cdots & w_{LQ'} \end{bmatrix} \begin{bmatrix} f_1(\alpha_0) \\ f_1(\alpha_1) \\ \vdots \\ f_1(\alpha_l) \\ \vdots \\ f_1(\alpha_{L'}) \\ 1 \end{bmatrix}$$

$$\boldsymbol{\beta} = \boldsymbol{w}\boldsymbol{b}$$

(Q by 1)

$$\hat{\boldsymbol{y}} = \begin{bmatrix} \hat{y}_0 \\ \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_q \\ \vdots \\ \hat{y}_{Q-1} \end{bmatrix} = \begin{bmatrix} f_2(\beta_0) \\ f_2(\beta_1) \\ f_2(\beta_2) \\ \vdots \\ f_2(\beta_l) \\ \vdots \\ f_2(\beta_{Q-1}) \end{bmatrix}$$

Two-Layer Neural Network의 일반화

- **N개의 데이터**

데이터 번호	입력	출력
	$x_0, \cdots, x_m, \cdots, x_{M-1}$	$y_0, \cdots, y_q, \cdots, y_{Q-1}$
0	$x_{00}, \cdots, x_{m0}, \cdots, x_{(M-1)0}$	$y_{00}, \cdots, y_{q0}, \cdots, y_{(Q-1)0}$
⋮	⋮	⋮
n	$x_{0n}, \cdots, x_{mn}, \cdots, x_{(M-1)n}$	$y_{0n}, \cdots, y_{qn}, \cdots, y_{(Q-1)n}$
⋮	⋮	⋮
$N-1$	$x_{0(N-1)}, \cdots, x_{m(N-1)}, \cdots, x_{(M-1)(N-1)}$	$y_{0(N-1)}, \cdots, y_{q(N-1)}, \cdots, y_{(Q-1)(N-1)}$

❖ **n번째 데이터에 대해서 (n은 0부터 N-1까지의 정수)**

α (L by 1) $\quad$ v (L by M+1) → weight = machine이라 볼 수 있는데, 데이터가 달라지는 거지 Machine이 달라지는 게 아니기 때문에 weight에는 n이 안붙음

$$\begin{bmatrix} \alpha_{0n} \\ \alpha_{1n} \\ \vdots \\ \alpha_{ln} \\ \vdots \\ \alpha_{L'n} \end{bmatrix} = \begin{bmatrix} v_{00} & v_{10} & \cdots & v_{M0} \\ v_{01} & v_{11} & \cdots & v_{M1} \\ \vdots & & \ddots & \vdots \\ v_{0l} & v_{1l} & \ddots & v_{Ml} \\ \vdots & & \ddots & \vdots \\ v_{0L'} & v_{1L'} & \cdots & v_{ML'} \end{bmatrix} \begin{bmatrix} x_{0n} \\ x_{1n} \\ \vdots \\ x_{Mn} \end{bmatrix}$$

x (M+1 by 1)

weight는 update 할때만 바뀜

❖ *Training 관점: n-1 번째 데이터로부터 업데이트 된 weights*

Neural Network 모델의 학습 알고리즘

- **Error Backpropagation (오차 역전파) 알고리즘**
  - 경사하강법 기반 → 지금까지의 경사하강이랑은 좀 다름
  - 개별 훈련 데이터에 대해서 알고리즘 적용
  - 설정한 Cost Function로부터 모델의 예측값과 실측값 간의 Error 값으로부터 역으로 전파되면서 Weight를 업데이트함
    - ✓ 본 강의자료에서는 Activation Function은 Sigmoid, Cost Function은 MSE 기준으로 설명

기존의 Gradient Descent (경사하강): $W_{new} = W_{old} - \alpha \frac{\partial f_{cost}}{\partial W_{old}}$ batch size: N

"모든" 데이터의 평균

뉴 - GD: SGD: $W_{new} = W_{old} - \alpha \frac{\partial f_{cost}}{\partial W_{old}}$ batch size: 1

→ 이래서 나온 개념

데이터 1개만

Stochastic

Batch Size

ex) 데이터 100개

GD: Update 1번 (모든 데이터의 평균으로 업뎃)

SGD: Update 100번 (각 데이터마다 업뎃)

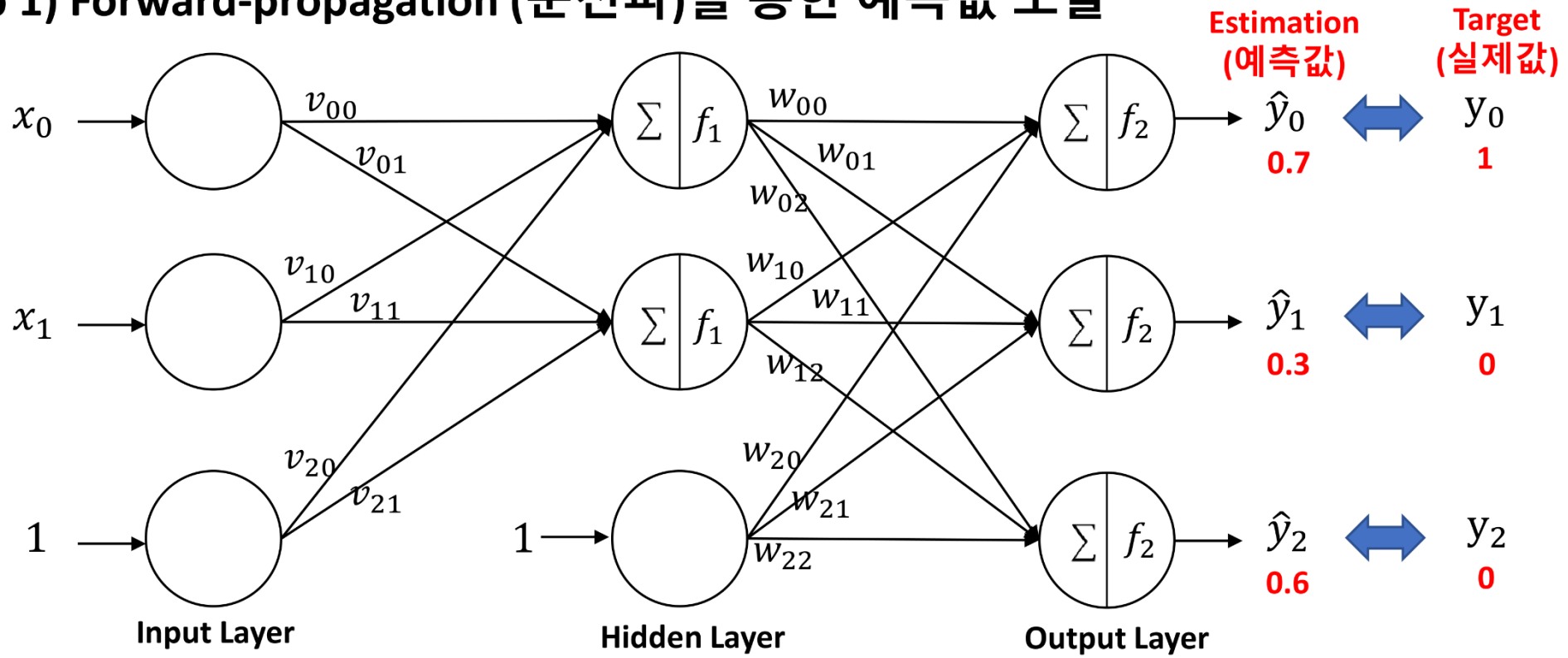
ex) 평균할 데이터 개수

2^n개로 잡는게 국룰

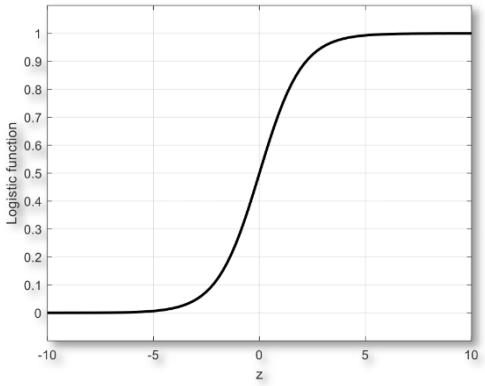
GPU의 연산 유닛이 2^n개이기 때문

Error Back-Propagation 알고리즘

Step 1) Forward-propagation (순전파)를 통한 예측값 도출



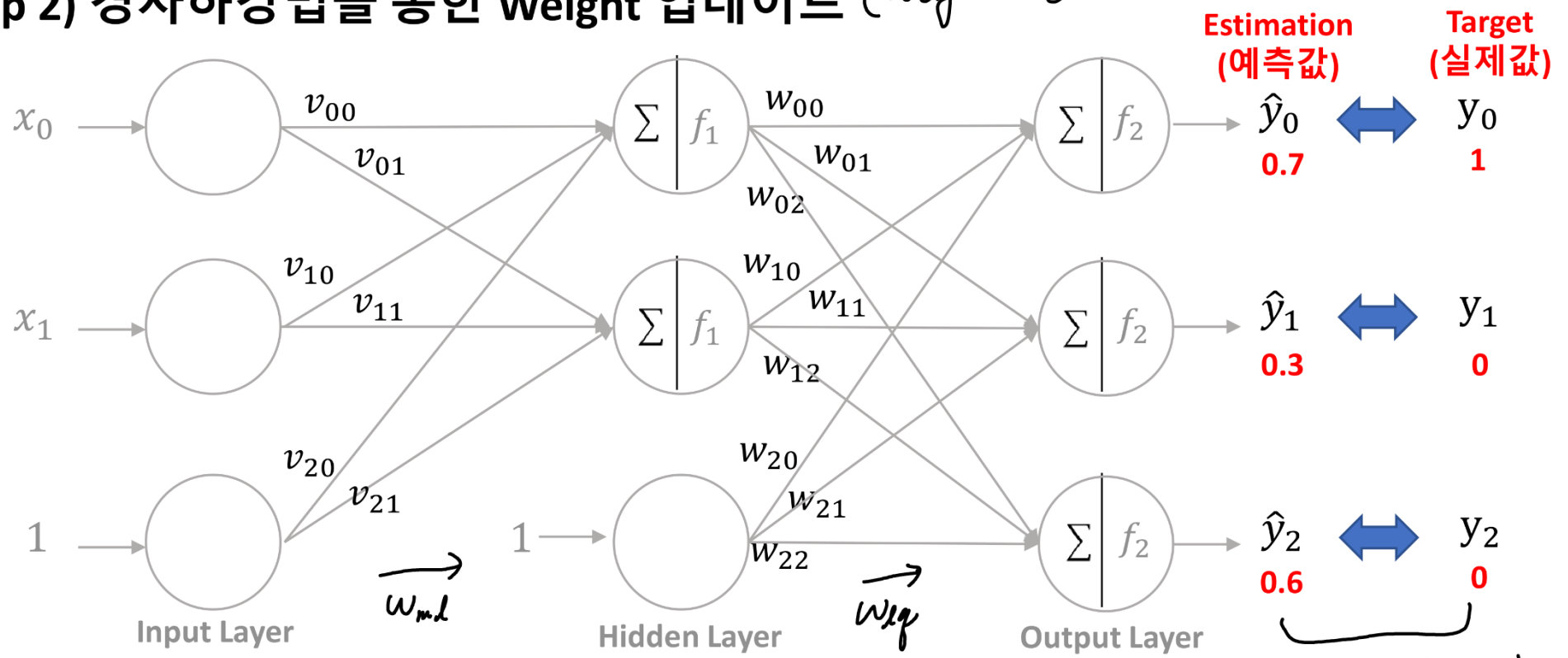
활성화함수 f_1과 f_2는 *Sigmoid 함수로 선택하였다고 가정*



- 예측값 계산 과정: Forward-propagation (순전파)
- Weight 업데이트 과정: Back-propagation(역전파)

Error Back-Propagation 알고리즘

Step 2) 경사하강법을 통한 Weight 업데이트 (Weight 개별적 업뎃)



목표: 이 둘의 차이를 최소화

❖ **Cost Function을 MSE로 설정하였을 때, n번째 데이터에 대해서**

$$v_{ml} \leftarrow v_{ml} - \eta \frac{\partial}{\partial v_{ml}} \epsilon_{MSE}(n)$$

$$w_{lq} \leftarrow w_{lq} - \eta \frac{\partial}{\partial w_{lq}} \epsilon_{MSE}(n)$$

$$\epsilon_{MSE}(n) = \sum_{q=0}^{Q-1} \left(\hat{y}_{qn} - y_{qn}\right)^2$$

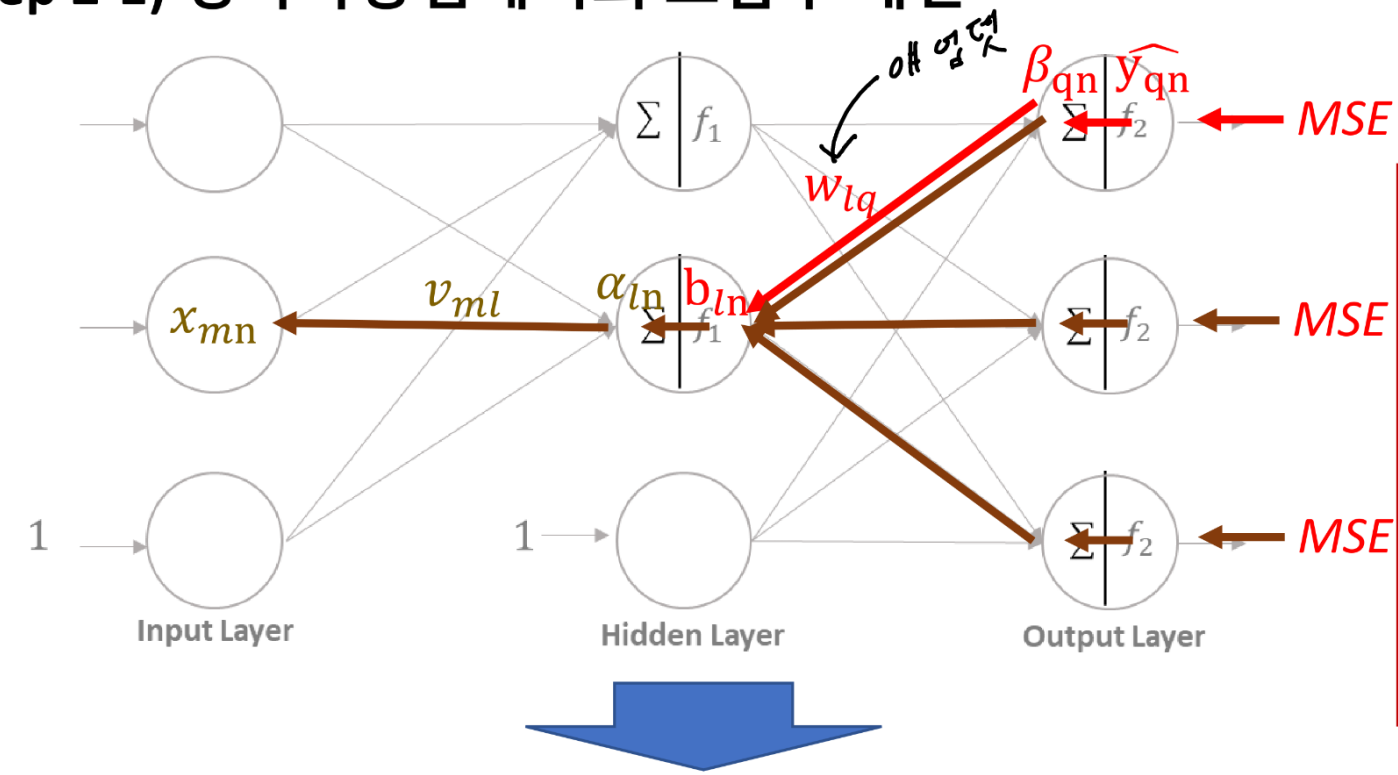
(미분해야되서 절댓값 대신 제곱)

Q 업데이트 Learning Rate 에서 조절가능

일단, 특정 weight에 대한 편미분 고려

Error Back-Propagation 알고리즘

Step 2-1) 경사하강법에서의 도함수 계산



Forward-propagation 에 사용한 weight (Old)

$$v_{ml}^{New} \leftarrow v_{ml}^{Old} - \eta \frac{\partial}{\partial v_{ml}} \epsilon_{MSE}(n)\Big|_{Old}$$

$$w_{lq}^{New} \leftarrow w_{lq}^{Old} - \eta \frac{\partial}{\partial w_{lq}} \epsilon_{MSE}(n)\Big|_{Old}$$

(합성함수 미분)

Sigmoid 미분 ← $\hat{y}(1-\hat{y})$

$$\frac{\partial}{\partial w_{lq}} \epsilon_{MSE}(n) = \frac{\partial \epsilon_{MSE}(n)}{\partial \hat{y}_{qn}} \cdot \frac{\partial \hat{y}_{qn}}{\partial \beta_{qn}} \cdot \frac{\partial \beta_{qn}}{\partial w_{lq}}$$

$$\frac{\partial}{\partial v_{ml}} \epsilon_{MSE}(n) = \frac{\partial \epsilon_{MSE}(n)}{\partial b_{ln}} \cdot \frac{\partial b_{ln}}{\partial \alpha_{ln}} \cdot \frac{\partial \alpha_{ln}}{\partial v_{ml}}$$

Error Back-Propagation 알고리즘

- $\frac{\partial}{\partial w_{lq}}\epsilon_{MSE}(n)$ 구하기

$$\frac{\partial}{\partial w_{lq}}\epsilon_{MSE}(n) = \frac{\partial \epsilon_{MSE}(n)}{\partial \hat{y}_{qn}} \cdot \frac{\partial \hat{y}_{qn}}{\partial \beta_{qn}} \cdot \frac{\partial \beta_{qn}}{\partial w_{lq}}$$

1 항 2 항 3 항

우변 1항

$$\frac{\partial \epsilon_{MSE}(n)}{\partial \hat{y}_{qn}} = \frac{\partial}{\partial \hat{y}_{qn}} \sum_{j=0}^{Q-1} (\hat{y}_{jn} - y_{jn})^2$$

$$= \frac{\partial}{\partial \hat{y}_{qn}} \left\{ (\hat{y}_{0n} - y_{0n})^2 + \cdots + (\hat{y}_{qn} - y_{qn})^2 + \cdots + (\hat{y}_{(Q-1)n} - y_{(Q-1)n})^2 \right\}$$

$$= 2(\hat{y}_{qn} - y_{qn})$$

우변 2항

$$\frac{\partial \hat{y}_{qn}}{\partial \beta_{qn}} = \frac{\partial}{\partial \beta_{qn}} f(\beta_{qn}) = f(\beta_{qn})\left(1 - f(\beta_{qn})\right) = \hat{y}_{qn}(1 - \hat{y}_{qn})$$

(참고) 시그모이드 함수의 미분

$$f'(x) = f(x)(1 - f(x))$$

Error Back-Propagation 알고리즘

- $\frac{\partial}{\partial w_{lq}} \epsilon_{MSE}(n)$ 구하기

$$\frac{\partial}{\partial w_{lq}} \epsilon_{MSE}(n) = \frac{\partial \epsilon_{MSE}(n)}{\partial \hat{y}_{qn}} \cdot \frac{\partial \hat{y}_{qn}}{\partial \beta_{qn}} \cdot \frac{\partial \beta_{qn}}{\partial w_{lq}}$$

우변 3항

$$\frac{\partial \beta_{qn}}{\partial w_{lq}} = \frac{\partial}{\partial w_{lq}} \sum_{j=0}^{L} w_{jq} b_{jn} = \frac{\partial}{\partial w_{lq}} \left(w_{0q} b_{0n} + \cdots + w_{lq} b_{ln} + \cdots w_{Lq} b_{Ln} \right) = b_{ln}$$

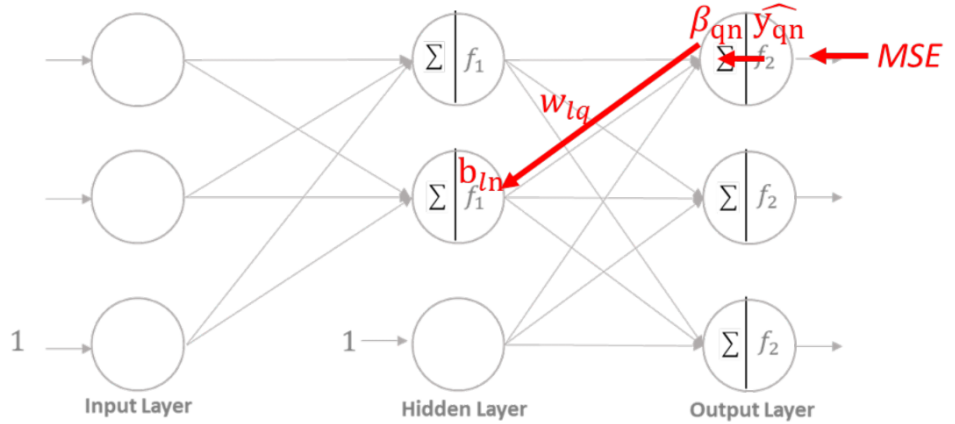
최종

$$\frac{\partial \epsilon_{MSE}(n)}{\partial w_{lq}} = \frac{\partial \epsilon_{MSE}(n)}{\partial \hat{y}_{qn}} \cdot \frac{\partial \hat{y}_{qn}}{\partial \beta_{qn}} \cdot \frac{\partial \beta_{qn}}{\partial w_{lq}}$$

$$= 2\left(\hat{y}_{qn} - y_{qn}\right) \hat{y}_{qn}\left(1 - \hat{y}_{qn}\right) b_{ln}$$

$$= \delta_{qn} b_{ln}$$

δ_{qn}: q번째 output node에 대해,
2*output*(output - target)*(1-output)



- *Forward-propagation 기준*
  - w_{lq} *통과* ***전*** *입력값 =>* b_{ln}
  - w_{lq} *통과* ***후*** *지나게 되는 모든 Node*
    => 출력층 q 번째 Node

Error Back-Propagation 알고리즘

- $\frac{\partial}{\partial v_{ml}}\epsilon_{MSE}(n)$ 구하기

Linear Combination

$$\frac{\partial}{\partial v_{ml}}\epsilon_{MSE}(n) = \frac{\partial \epsilon_{MSE}(n)}{\partial b_{ln}} \cdot \frac{\partial b_{ln}}{\partial \alpha_{ln}} \cdot \frac{\partial \alpha_{ln}}{\partial v_{ml}}$$

우변 1항

$$\frac{\partial \epsilon_{MSE}}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial \beta} \times \frac{\partial \beta}{\partial b}$$

Sigmoid 미분

$$\frac{\partial \epsilon_{MSE}(n)}{\partial b_{ln}} = \frac{\partial}{\partial b_{ln}}\sum_{q=0}^{Q-1}(\hat{y}_{qn}-y_{qn})^2 = 2\sum_{q=0}^{Q-1}(\hat{y}_{qn}-y_{qn})\frac{\partial \hat{y}_{qn}}{\partial b_{ln}}$$

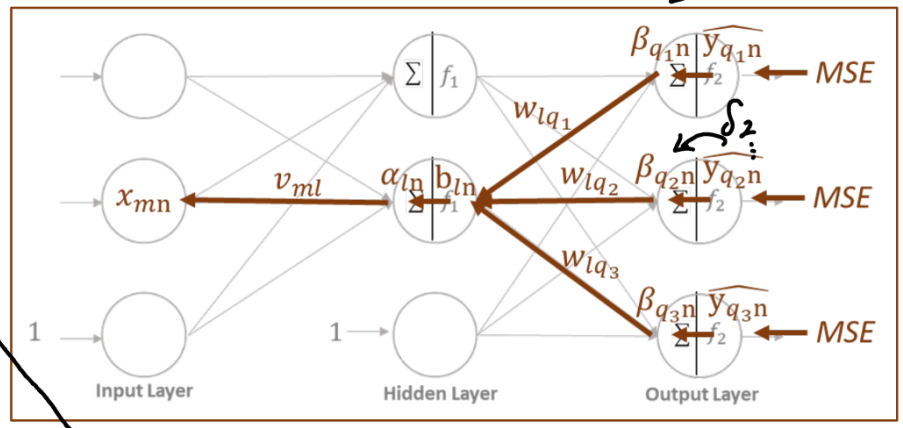
$$= 2\sum_{q=0}^{Q-1}(\hat{y}_{qn}-y_{qn})\frac{\partial}{\partial b_{ln}} f\left(\sum_{j=0}^{L} w_{jq}b_{jn}\right)$$

$$= 2\sum_{q=0}^{Q-1}(\hat{y}_{qn}-y_{qn})\frac{\partial}{\partial b_{ln}} f(w_{0q}b_{0n}+\cdots+w_{lq}b_{ln}+\cdots+w_{Lq}b_{Ln})$$

$$= 2\sum_{q=0}^{Q-1}(\hat{y}_{qn}-y_{qn}) f\left(\sum_{j=0}^{L} w_{jq}b_{jn}\right)\left(1-f\left(\sum_{j=0}^{L} w_{jq}b_{jn}\right)\right)w_{lq}$$

$$= 2\sum_{q=0}^{Q-1}(\hat{y}_{qn}-y_{qn})\,\hat{y}_{qn}(1-\hat{y}_{qn})w_{lq}$$

$$= \sum_{q=0}^{Q-1}\delta_{qn}\,w_{lq}$$



V는 모든 노드에 영향을 받아서 Σ를 불여야됨

l 번째 Hidden Node와 연결된 weight들과 Output Node를 고려하게 됨.

Error Back-Propagation 알고리즘

- $\frac{\partial}{\partial v_{ml}}\epsilon_{MSE}(n)$ 구하기

$$\frac{\partial}{\partial v_{ml}}\epsilon_{MSE}(n) = \frac{\partial \epsilon_{MSE}(n)}{\partial b_{ln}} \cdot \frac{\partial b_{ln}}{\partial \alpha_{ln}} \cdot \frac{\partial \alpha_{ln}}{\partial v_{ml}}$$

우변 2항

$$\frac{\partial b_{ln}}{\partial \alpha_{ln}} = \frac{\partial}{\partial \alpha_{ln}} f(\alpha_{ln}) = f(\alpha_{ln})\big(1 - f(\alpha_{ln})\big) = b_{ln}(1 - b_{ln})$$

Sigmoid 미분

우변 3항

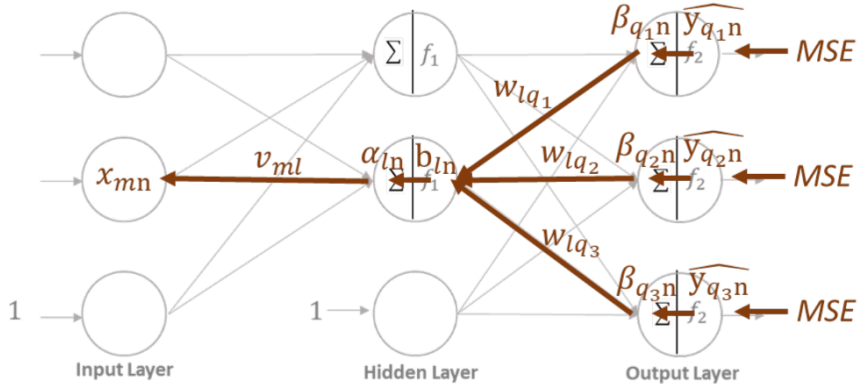
$$\frac{\partial \alpha_{ln}}{\partial v_{ml}} = \frac{\partial}{\partial v_{ml}} \sum_{j=0}^{M} v_{jl} x_{jn} = \frac{\partial}{\partial v_{ml}} (\cancel{v_{0l} x_{0n}} + \cdots + v_{ml} x_{mn} + \cdots + \cancel{v_{Ml} x_{Mn}}) = x_{mn}$$

(편미분이라)

최종

$$\frac{\partial \epsilon_{MSE}(n)}{\partial v_{ml}} = \frac{\partial \epsilon_{MSE}(n)}{\partial b_{ln}} \cdot \frac{\partial b_{ln}}{\partial \alpha_{ln}} \cdot \frac{\partial \alpha_{ln}}{\partial v_{ml}}$$

$$= \sum_{q=0}^{Q-1} \delta_{qn} w_{lq}\, b_{ln}(1 - b_{ln}) x_{mn}$$



- v_{ml} 통과 **전** 입력값 => x_{mn}
- v_{ml} 통과 **후** 지나게 되는 모든 Node

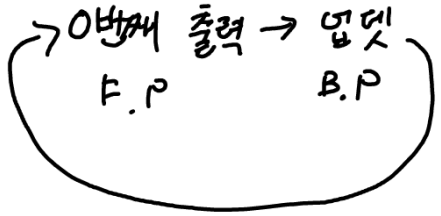
=> l 번째 Hidden node, 모든 Output Node

Neural Network 모델의 학습 알고리즘

- **전체 알고리즘 적용 순서**

1) 신경망 모델 설계 (Hidden Layer 수, Node 수, 활성화 함수, Learning Rate 등)
2) 설계된 모델에 따른 Weight Matrix 생성 및 초기화 (랜덤으로 Weight 넣고 Forward Propagation)
   Weight는 음수가능
3) N개의 훈련 데이터 Shuffle
4) 0부터 N-1번째 데이터에 대해 순차적으로 오차 역전파 알고리즘 적용 (1 epoch)
   - **for** n번째 데이터 모든 데이터를 다 하면 1epoch
     - (순전파) n-1에서 업데이트 된 Weight로부터 예측값 $\hat{y}_{qn}$ 계산
     - (역전파) 실제값(Target)과 예측값(Output) 간의 오차로부터 Weight

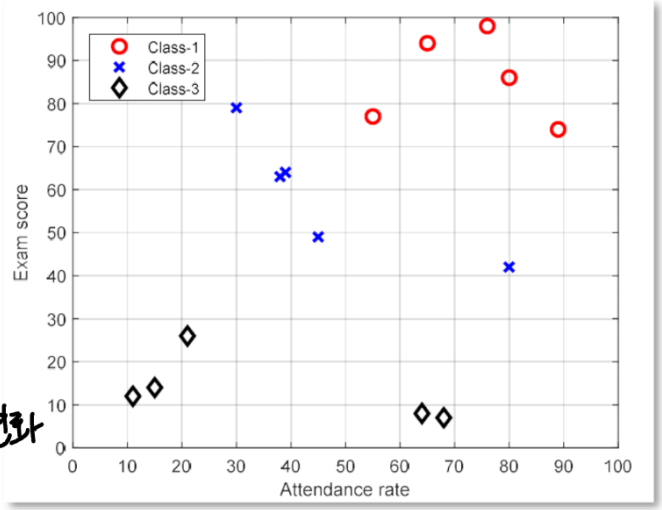
   ~~X업데이트 => 주의사항: Input layer에 가까운 weight부터 업데이트~~
5) 업데이트 된 weight로부터 accuracy 계산
6) 사용자가 입력한 epoch 수 만큼 3~4) 반복



Two-Layer Neural Network 적용 예시

신경망 손으로 푸는 문제는 반드시 나옴

- 데이터
  - 2입력 3클래스 분류 데이터
    - 입력 속성: 출석률, 시험성적
    - 출력 속성: 학점 (A, B, C)



☆ 손으로: 최기 weight 후 F.P 후 업뎃 후 weight 변화 +Confusion Matrix 까지

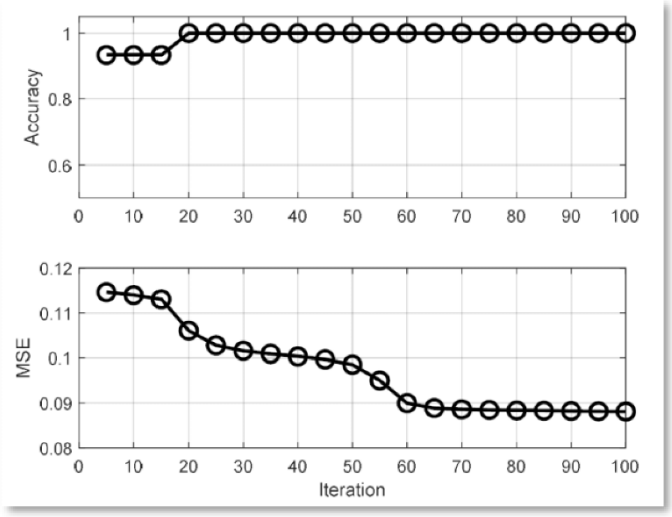
데이터 번호	입력		출력
	출석률	시험성적	학점
0	76	98	A
1	65	94	A
2	80	86	A
3	89	74	A
4	55	77	A
5	30	79	B
6	39	64	B
7	38	63	B
8	45	49	B
9	80	42	B
10	68	7	C
11	64	8	C
12	21	26	C
13	15	14	C
14	11	12	C

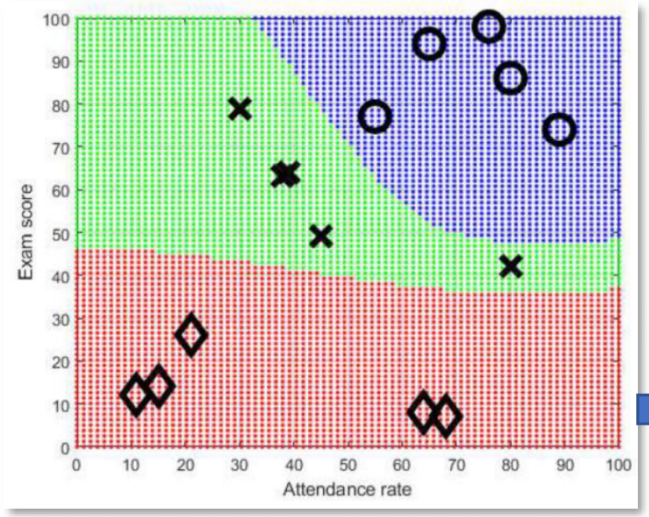
One-Hot Encoding

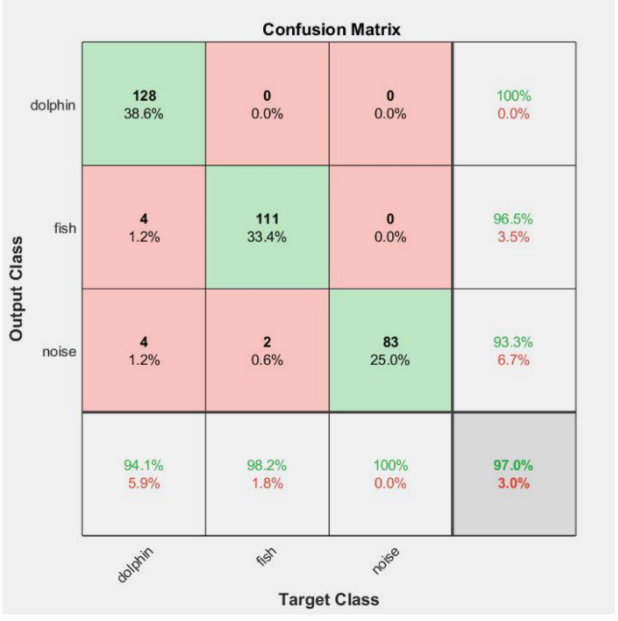
데이터 번호	입력		출력		
	출석률	시험성적	y_0	y_1	y_2
0	76	98	1	0	0
1	65	94	1	0	0
2	80	86	1	0	0
3	89	74	1	0	0
4	55	77	1	0	0
5	30	79	0	1	0
6	39	64	0	1	0
7	38	63	0	1	0
8	45	49	0	1	0
9	80	42	0	1	0
10	68	7	0	0	1
11	64	8	0	0	1
12	21	26	0	0	1
13	15	14	0	0	1
14	11	12	0	0	1

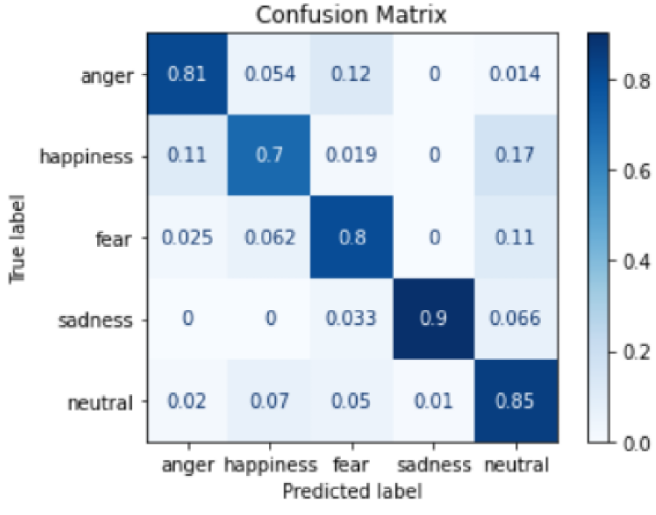
Two-Layer Neural Network 적용 예시

- 학습 성공 시,



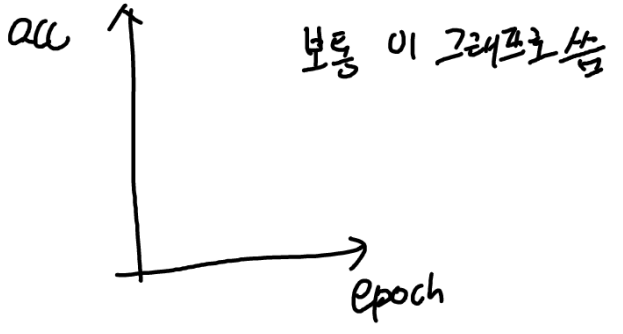






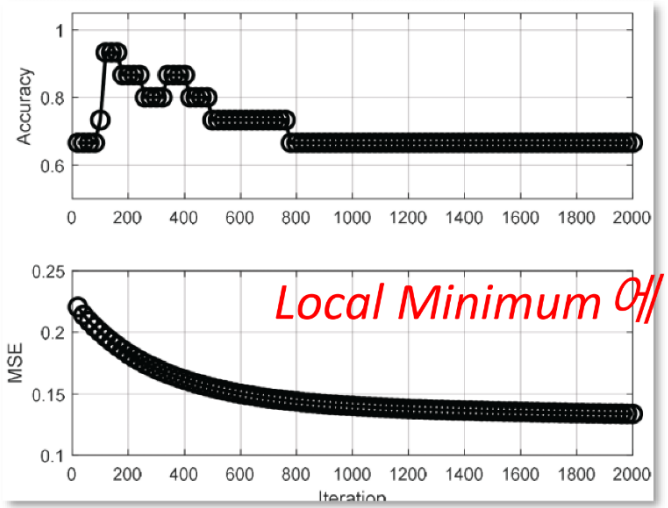
Confusion Matrix (모델 성능을 표현하는 가장 보편적 방법)

Confusion Matrix로 나타내 보자

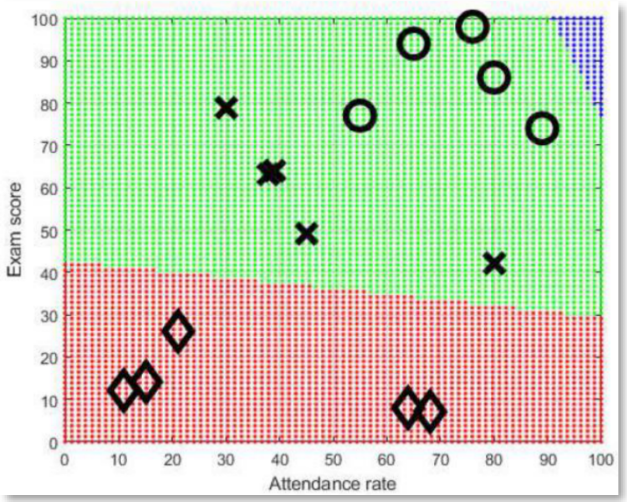


Two-Layer Neural Network 적용 예시

- 학습 실패 시,



Local Minimum 에 빠짐



위의 결과를 Confusion Matrix로 나타내 보자

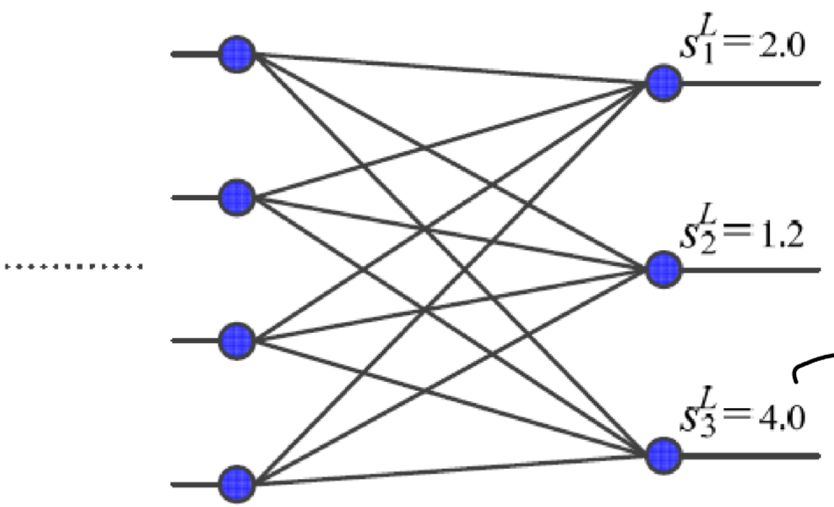
Softmax Function

지능형 예측·진단 연구실
Intelligent Prognostics & Diagnostics Lab.

- **다중 분류를 위한 Softmax Function**

$$o_j = \frac{e^{s_j}}{\sum_{i=1,c} e^{s_i}}$$

- 동작 예시
  - softmax는 max를 모방(출력 노드의 중간 계산 결과 s_i^L에서 최댓값은 더욱 활성화하고 작은 값은 억제
  - 모두 더하면 1이 되어 확률 모방



로지스틱 시그모이드	max	softmax
0.8808	0.0	0.1131
0.7685	0.0	0.0508
0.9820	1.0	0.8360

Softmax Function

- **Softmax를 출력함수로 사용할 때의 Cost Function**
  - **Cross-Entropy**

❖ 이진분류 =>

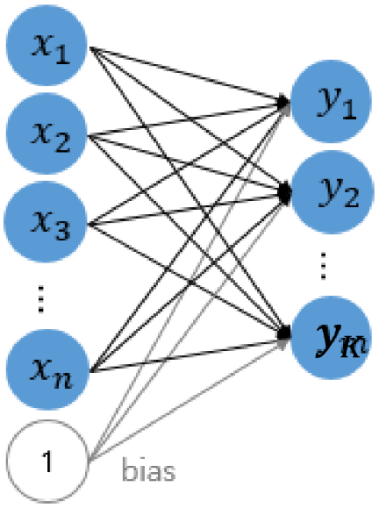
$$\epsilon_{CEE} = -\frac{1}{N}\sum_{n=0}^{N-1}\{y_n \ln p_n + (1-y_n)\ln(1-p_n)\}$$

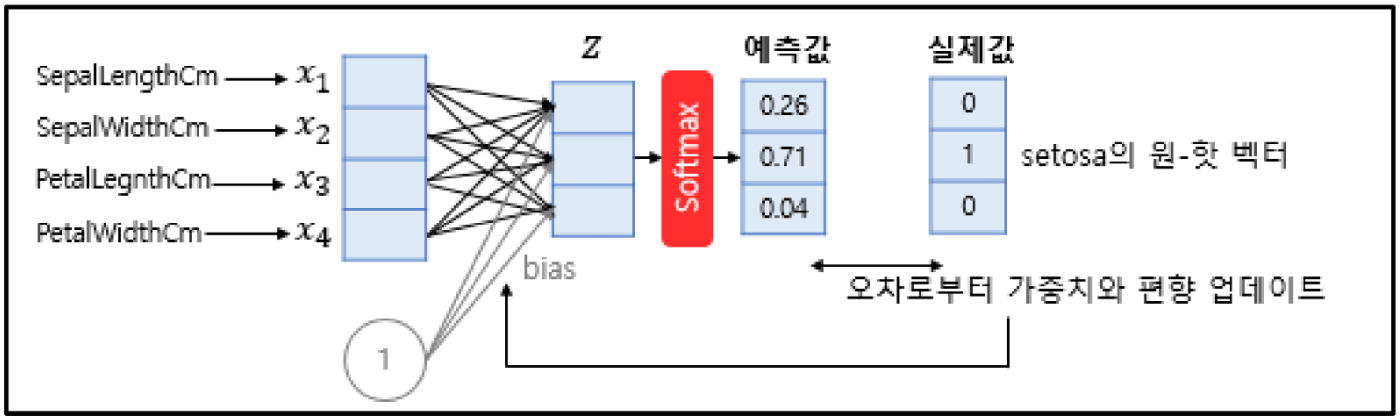
$$\{y_n \ln p_n + (1-y_n)\ln(1-p_n)\}$$

$$= \{y_{0,n} \ln p_{0,n} + (y_{1,n})\ln(p_{1,n})\}$$

$$= \sum_{k=0}^{1} y_{k,n} ln p_{k,n} \Rightarrow \sum_{k=0}^{K-1} y_{k,n}\ln(p_{k,n})$$

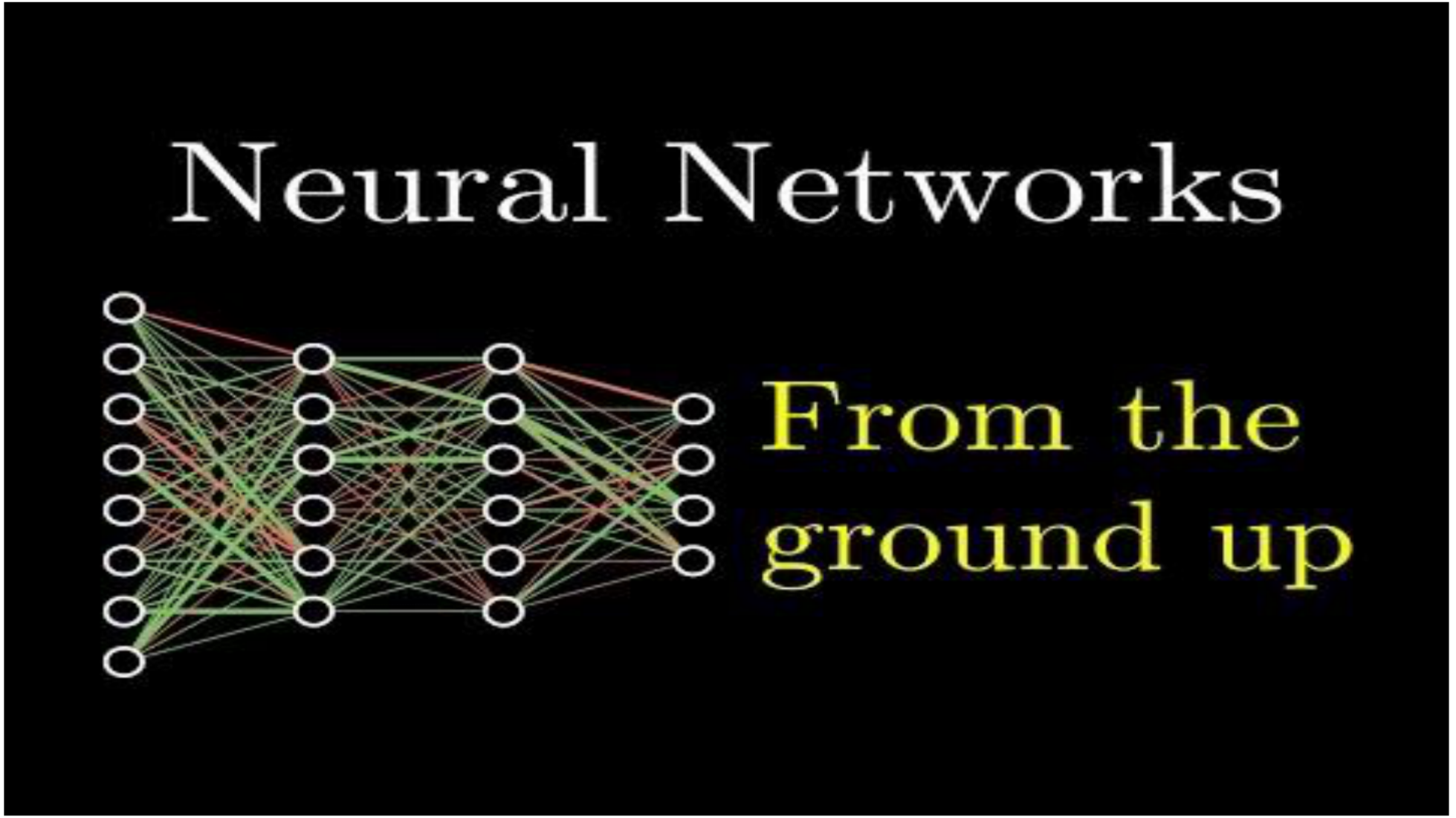
$$\epsilon_{CEE} = -\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=0}^{K-1} y_{i,n}\ln(p_{i,n})$$





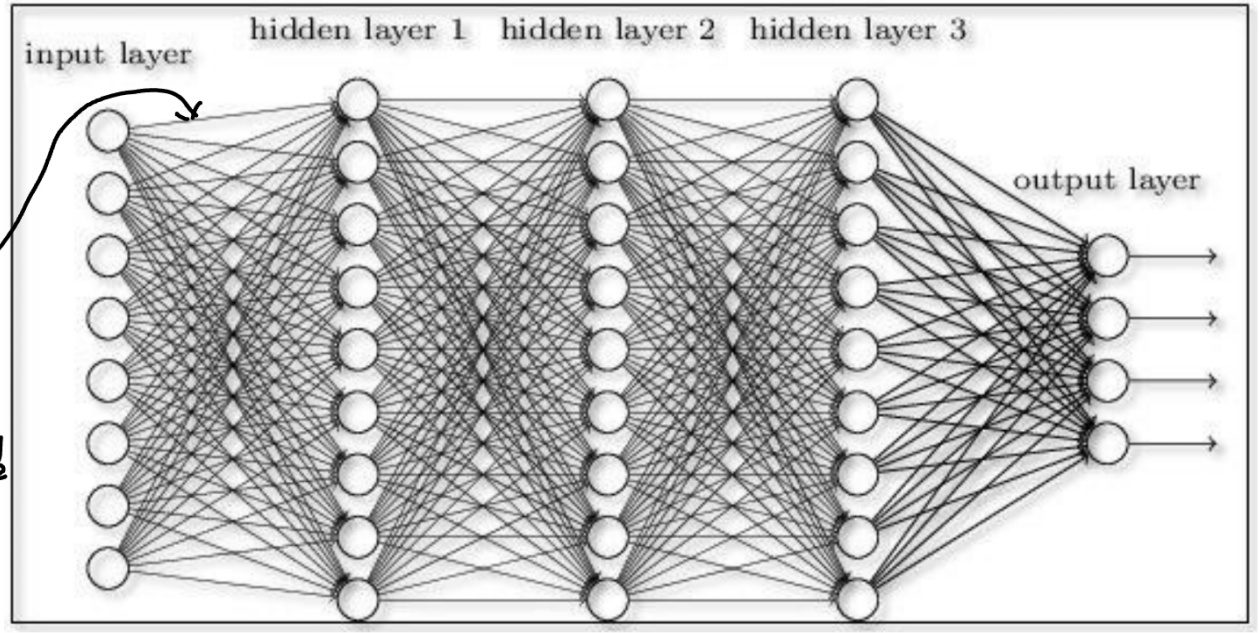
ANN 정리

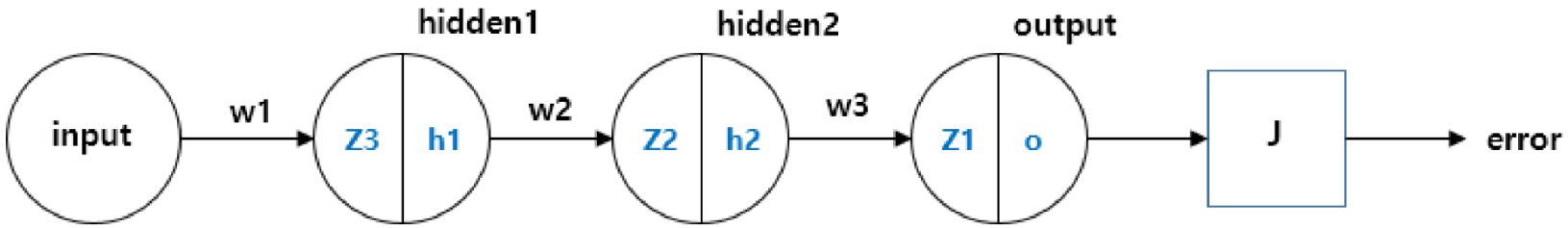
https://www.youtube.com/watch?v=aircAruvnKk



Deep Learning으로의 확장

$f(z)$가 0~1일때
$f(z)(1-f(z))$
는 0~0.25
→Hidden Layer가
많아지면
앞쪽
업뎃할때 값이 점점 작아짐
→ Gradient Vanishing

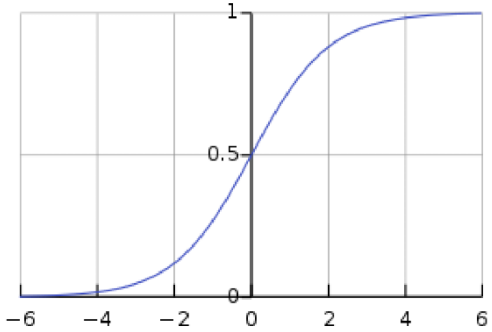




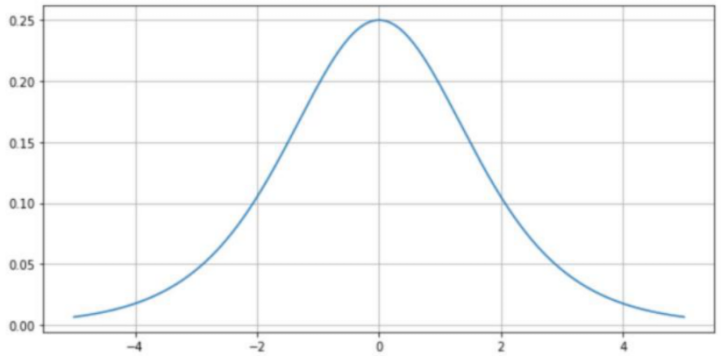
Node의 활성화 함수가 모두 동일하다고 했을 때,
w1을 update하기 위해서는 활성화 함수의 도함수를 3번 곱한 값이 들어감.

Deep Learning으로의 확장

Sigmoid 함수



Sigmoid의 도함수



도함수를 곱할수록 값이 작아져
Weight Update가 잘 이뤄지지 않음.
=> Hidden Layer가 많을수록 불리

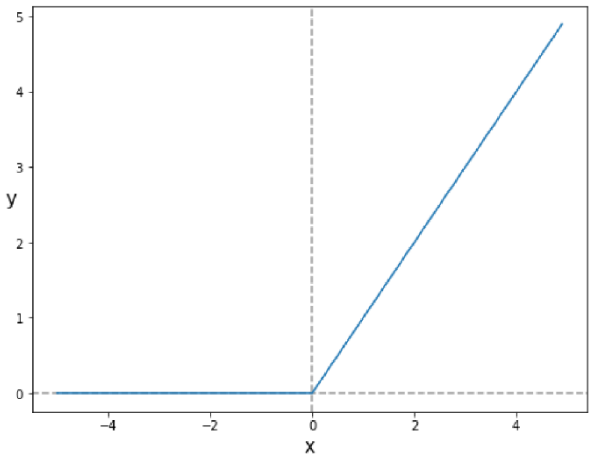
"Gradient Vanishing Problem"

Leacky ReLU 애같은거도 있음

ReLU 함수

Gradient Vanishing 문제를 해결하기 위해 제안된 활성화 함수

ReLU

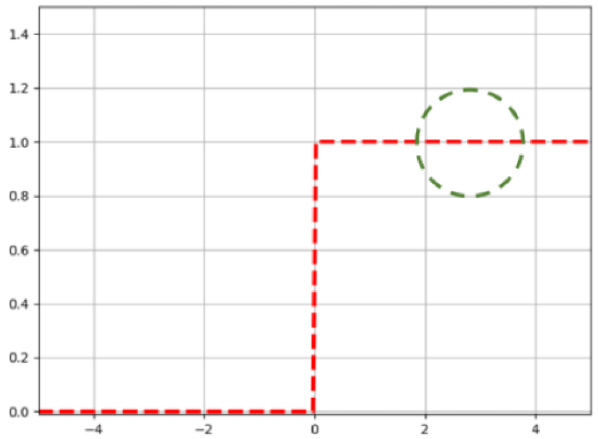


미분은 안되지만 코드 쳐보니까 잘됨

애는 선형이라 그냥 Logistic

$$f(x) = \begin{cases} 0 & x < 0 \\ x & x \geq 0 \end{cases} \qquad f(x) = max(0, x)$$

dReLU(x)/dx

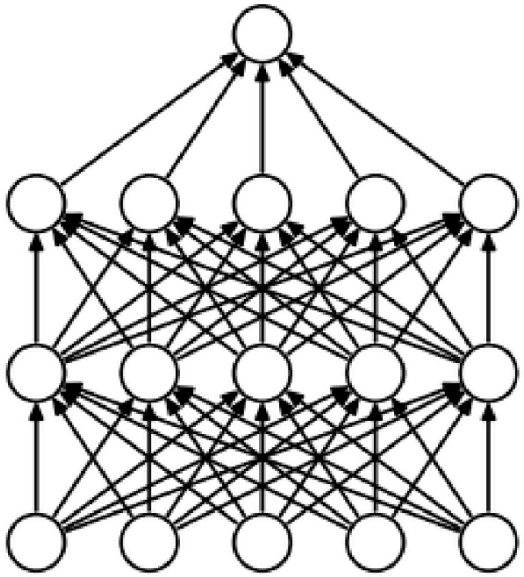


Hidden Layer가 많아도,
즉, Neural Network가 Deep해져도
Weight를 업데이트 할 수 있음

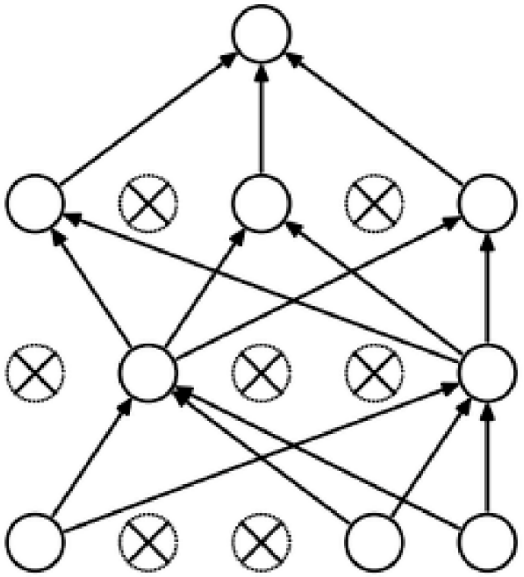
Dropout (고급 머신러닝에서)

- **Dropout (드롭아웃)**
  - 신경망의 일부 Node를 랜덤으로 비활성화하며 훈련시키는 기법
  - Regularization (Overfitting을 방지하기 위해 사용)

일반화 성능 ↑



(a) Standard Neural Net



(b) After applying dropout.

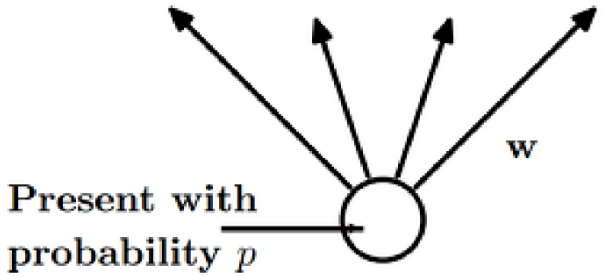
각 node가 존재할 확률 p

통상 알려진 default 입력층: 0.8 은닉층: 0.5

출력층은?

Dropout

- **Dropout에서의 Training 단계**



(a) At training time

l번째 은닉층의 j번째 노드의 연산:

$$z_j^l = \tau_l(s_j^l)$$

이때 $s_j^l = \mathbf{u}_j^l \mathbf{z}^{l-1}$

$\Rightarrow$

드롭아웃 적용:

$$z_j^l = \tau_l(s_j^l)$$

이때 $\begin{cases} \tilde{\mathbf{z}}^{l-1} = \mathbf{z}^{l-1} \odot \boldsymbol{\pi}^{l-1} \\ s_j^l = \mathbf{u}_j^l \tilde{\mathbf{z}}^{l-1} \end{cases}$

Boolean 배열 $\boldsymbol{\pi}$에 노드 제거 여부를 표시

Boolean 배열: 배열 내부에 True or False만 존재

Dropout

- **Dropout에서의 Test 단계**
  - 가중치에 생존 비율(p)을 곱하여 전방 계산
  - 학습 과정에서 가중치가 생존 비율(p) 만큼만 참여했기 때문